Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/01/2005
PI Name:	Plaut, Karen Ph.D.		
Project Title:	Is Gravity A Continuum? Investigations Of Mechanisms Mediating Gravity-Induced Change In Mammary Metabolism And Maternal Behavior In Pregnant And Lactating Rats.		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	01/01/2005	**End Date:**	07/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	5	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 7/31/07, per A. Chu (7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wade, Charles (United States Army Institute for Surgical Research) Ronca, April (Wake Forest University School of Medicine)		
Grant/Contract No.:	NNA05CP91A ; NCC2-1373		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Homeostasis, the dynamic state of physiological balance, is fundamentally changed in mammals exposed to either microgravity (spaceflight) or hypergravity (centrifugation). In pregnant and lactating mammals, failure to maintain homeostasis exerts pronounced effects on both mothers and their offspring. Our previous data, derived from pregnant rats exposed to either micro or hypergravity, support: 1) decreased mammary metabolic activity with increased gravitational force; 2) changes in maternal and neonatal energy balance; 3) altered maternal care; and 4) neonatal mortality. Gestation day 11 (G11) rats were placed on the 24 ft. centrifuge and exposed to 2-g hypergravity (HG), remained at 1-g as stationary controls (SC) or were pair-fed (PF) to match feed intake of the HG group. To determine metabolic activity, mammary tissue was collected from anesthetized animals, sliced 0.5 mm thick and placed in flasks containing Krebs-Ringer Buffer with 10mM glucose, 1 microg/ml insulin and 1 microCi/flask of U-14C labeled glucose. The flasks were gassed with O2, sealed with rubber stoppers containing filter paper and incubated at 37C for 3 hrs, in a shaking water bath. Metabolic activity was stopped by killing the tissue with 0.5 M H2SO4. Glucose oxidation to CO2 was determined by counting 14C-CO2 trapped on the filter papers. CO2 incorporation into lipid was measured by petroleum ether extractions and measuring in a liquid scintillation counter. Results were expressed as nmoles of U-14C glucose utilized per 100 mg tissue/3 hrs. Mammary metabolism as measured by glucose oxidation to carbon dioxide and incorporation into lipid was significantly decreased in response to hypergravity during pregnancy and lactation. 1.)To measure the effects of prolactin on mammary metabolism of pregnant and lactating rates 2.) To measure behavioral changes that are coupled with physiological changes when animals are exposed to prolactin during hypergravity and have the additional stress of pregnancy and lactation. 3.) To determine whether food intake plays a role in the changes observed in hypergravity These tasks will help us to understand the physiological regulation of metabolism in altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have established a very complete data set that describes changes in metabolism in mammary tissue, liver and adipose (liver and adipose funded by NASA-EPSCoR) in animals exposed to hypergravity that are pregnant or lactating. In particular this data set can be used to examine the coordinated regulation of lipid metabolism and may provide insight into obesity related disorders as well as gestational diabetes.
Task Progress:	Subsequent studies on the centrifuge have allowed us to examine the effects of prolactin (Prl) , glucocorticoids and bromoergocryptine on mammary metabolism in animals exposed to hypergravity. Neither Prolactin nor bromocryptine injection affected mammary metabolism. That is, glucose oxidation to carbon dioxide was reduced about 30% and glucose incorporation into lipiid was reduced 70% compared to stationary controls. This was not different than the hypergravity controls. We also pair fed a group of animals to match the intake of animals exposed to hypergravity and found that changes in mammary metabolism were due to hypergravity and in general were independent of food intake. Expression of number of metabolic genes such as ATP citratate lyase and acetyl co A carboxylase as well as the Prl receptor were altered due to hypergravity indicating that changes in gene expression during hypergravity acccompany the observed metabolic changes. Further experiments are underway to determine the relationships among gene expression, metabolic rate, and hypergravity.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Zakrzewska E, Lintault L, Casey T, Maple R, Ronca A, Baer L, Wade C, Plaut K. "Can prolactin supplementation alleviate impaired metabolic activity of lactating rats exposed to hypergravity? " Annual Meeting of the American Society for Gravitational and Space Biology, Reno NV, November 2005. Gravit Space Biol. 2005 Nov;19(1):62. , Nov-2005
Abstracts for Journals and Proceedings	Lintault L, Zakrzewska E, Maple R, Baer L, Ronca A, Wade C, Plaut K. "Effects of hypergravity and food intake on gene expression of enzymes in the liver of periparturient rats." Annual Meeting of the American Society for Gravitational and Space Biology, Reno, NV, November 2005. Gravit Space Biol. 2005 Nov;19(1):20. , Nov-2005
Abstracts for Journals and Proceedings	Zakrzewska E, Lintault L, Maple R, Baer L, Ronca A, Wade C, Plaut K. "Effects of 2 x gravity (2-g) on prolactin receptor expression in adipose tissue of periparturient rats. " Experimental Biology 2005. FASEB Journal; 2005 Mar; abstract supplement. , Mar-2005
Dissertations and Theses	Lintault L. "EFFECTS OF HYPERGRAVITY AND FOOD INTAKE ON TISSUE METABOLISM, GENE EXPRESSION AND ENERGY BALANCE IN PERIPARTURIENT RATS." September 2005. , Sep-2005